



ADP EC Plan for Reflector

ADP GD team
2023/08/31

Reflector EC Requirement

- EC reason : Toray reflector material EOL (EOL in July'Y23)
- Action : ADP evaluates new films for replacement to avoid material and supply shortage.

Material comparison table

Group A	As is / To be	TCJ18N	EST188P	Model
TCJ18N EC to EST188P	Thickness(mm)	0.233	0.228	G270QAN01.2
	Optical Gain(%)	100%	100.2%	
	Chromaticity $\Delta(x,y)$	(0.0000,0.0000)	(-0.0001,-0.0001)	

RA test



RA Test Summary Report

保管單位 Storage	RA
保存年限 Retention Period	依產品適用範圍保存(參照： AUOQP-05-002 紀錄管制流程)

Product No	G270QAN01.3			Part No	97.27G03.302			Applicant/Dept		王秀芬/RRPBF3		Judgement	
Test Purpose	G270QAN01.3 New half tone 光罩cell + NSPZ					Application Date		2022/08/12		PASS			
RA No	D220816022		Product Stage	DVT		Input Date		2022/08/17					
Test Item	Test Condition	Test Panel				Test Result							
Hot-start	60°C 5	No	Chip_Id	Shipping_No		T0★	T1+F5★					Judgement	Judgement
		D2208160220001		xl2a5365qmzzzs0102		OK	OK					PASS	PASS
		D2208160220002		xl2a5365qfzzzs0102		OK	OK					PASS	
		D2208160220003		xl2a5365vdzzzs0102		OK	OK					PASS	
HTO	50°C 500h	No	Chip_Id	Shipping_No		T0★	T100★	T200★	T300★	T500		Judgement	Judgement
		D2208160220001		xl2a5365qmzzzs0102		OK	OK	OK	OK	OK		PASS	PASS
		D2208160220002		xl2a5365qfzzzs0102		OK	OK	OK	OK	OK		PASS	
		D2208160220003		xl2a5365vdzzzs0102		OK	OK	OK	OK	OK		PASS	
Cold-start	-20°C 5	No	Chip_Id	Shipping_No		T0★	T1+F5★					Judgement	Judgement
		D2208160220004		xl2a5350gbzzzs0102		OK	OK					PASS	PASS
		D2208160220005		xl2a5350ahzzzs0102		OK	OK					PASS	
		D2208160220006		xl2a5350akzzzs0102		OK	OK					PASS	

RA test



RA Test Summary Report

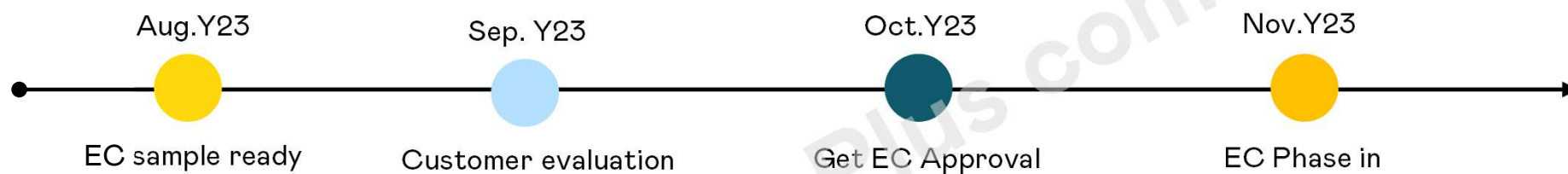
保管單位 Storage	RA
保存年限 Retention Period	依產品適用範圍保存(參照： AUOQP-05-002 紀錄管制流程)

Product No	G270QAN01.3		Part No	97.27G03.302		Applicant/Dept		王秀芬/RRPBF3		Judgement		
Test Purpose	G270QAN01.3 New half tone 光罩cell + NSPZ					Application Date		2022/08/12		PASS		
						Input Date		2022/08/17				
RA No	D220816022		Product Stage	DVT		Issue Date		2022/08/17				
Test Item	Test Condition	Test Panel				Test Result						
LTO	0°C 500h	No	Chip_Id	Shipping_No		T0★	T100★	T200★	T300★	T500	Judgement	Judgement
		D2208160220004		xl2a5350gbzzzs0102		OK	OK	OK	OK	OK	PASS	PASS
		D2208160220005		xl2a5350ahzzzs0102		OK	OK	OK	OK	OK	PASS	
		D2208160220006		xl2a5350akzzzs0102		OK	OK	OK	OK	OK	PASS	
THB	50°C 80% 500h	No	Chip_Id	Shipping_No		T0★	T100★	T200★	T300★	T500	Judgement	Judgement
		D2208160220007		xl2a5365vfzzzs0102		OK	OK	OK	OK	OK	PASS	PASS
		D2208160220008		xl2a5350ckzzzs0102		OK	OK	OK	OK	OK	PASS	
		D2208160220009		xl2a5350fbzzzs0102		OK	OK	OK	OK	OK	PASS	
TST	-20°C(30min) ~ 60°C(30min), 100cycles judge, 300cycles ref. 300cycles	No	Chip_Id	Shipping_No		C0★	C100★	C300			Judgement	Judgement
		D2208160220010		xl2a5365thzzzs0102		OK	OK	OK			PASS	PASS
		D2208160220011		xl2a53651kzzzs0102		OK	OK	OK			PASS	
		D2208160220012		xl2a53651hzzzs0102		OK	OK	OK			PASS	

Optical report

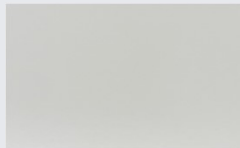
Optical Measurement Report													
Originator	chongfeng		Model/PCS		G270QAN01.3/5pcs				Date	2022/9/1			
Test Condition	Lum.,CR,Chromaticity,Uniformity,Gary Scale,V-T,View Angle,RT,CrossTalk												
Test System	Detector	Inverter			DC Voltage				Warm up time				
FPM-520-1	SR-UL1R	NA			24V				30 min				
【Test Reason】													
G270QAN01.3 New half tone 光罩cell + NSPZ													
【Judgement】													
PASS													
Test Items	unit	PSS			23.5°C					-4σ	mean	+4σ	
					2/8/31 上午 09:46 2/8/31 上午 10:28 2/8/31 下午 01:17 2/8/31 下午 02:03 2/8/31 下午 02:45								
					L2A5350GFZZS010L2A5350AFZZS010L2A53652FZZS010L2A53651MZZS010L2A5365VBZZS010								
		min	typ	max									
Center Brightness	cd/m ²	320	400		479.2	460.6	479.5	443.1	472.9	405.474	467.0	528.619	
Darkness (L0)					0.31	0.26	0.29	0.25	0.27	0.184	0.28	0.371	
Contrast Ratio(CR)	---	1280	1600		1529.0	1740.3	1678.9	1752.5	1744.2	1312.516	1689.0	2065.454	
Brightness(5P Ave.)	cd/m ²				460.1	445.9	461.7	428.9	461.5	394.314	451.6	508.940	
Chromaticity													
	Wx	---	0.280	0.310	0.340	0.299	0.298	0.304	0.303	0.298	0.290	0.301	0.312
	Wy	---	0.293	0.323	0.353	0.313	0.312	0.313	0.319	0.312	0.301	0.314	0.326
	Rx	---	0.659	0.689	0.719	0.687	0.687	0.683	0.687	0.687	0.679	0.686	0.693
	Ry	---	0.279	0.309	0.339	0.309	0.310	0.313	0.309	0.310	0.303	0.310	0.317
	Gx	---	0.181	0.211	0.241	0.213	0.213	0.213	0.213	0.213	0.212	0.213	0.213
	Gy	---	0.699	0.729	0.759	0.729	0.729	0.729	0.730	0.730	0.728	0.729	0.731
	Bx	---	0.117	0.147	0.177	0.149	0.149	0.148	0.149	0.149	0.147	0.149	0.150
	By	---	0.027	0.057	0.087	0.059	0.058	0.059	0.060	0.058	0.055	0.059	0.062
	NTSC	%				108.9	109.1	108.0	108.9	109.0	107.069	108.793	110.517
	Adobe			99		99.9%	99.9%	99.9%	99.9%	99.9%	0.998	0.999	1.000
	DCI-P3			98		98.4%	98.4%	98.4%	98.5%	98.4%	0.983	0.984	0.986
	CCT	---				7548	7629	7201	7214	7619	6574.847	7442.169	8309.491
Uniformity													
	5P					1.099	1.076	1.078	1.068	1.114	1.012	1.087	1.162
	9P	White		1.250	1.333	1.235	1.241	1.255	1.191	1.257	1.130	1.236	1.342
	13P					1.235	1.241	1.255	1.191	1.279	1.111	1.240	1.370
	9P Δx				0.010	0.003	0.005	0.005	0.005	0.006	0.001	0.005	0.009
	9P Δy				0.012	0.003	0.005	0.005	0.007	0.008	-0.003	0.005	0.014

EC plan

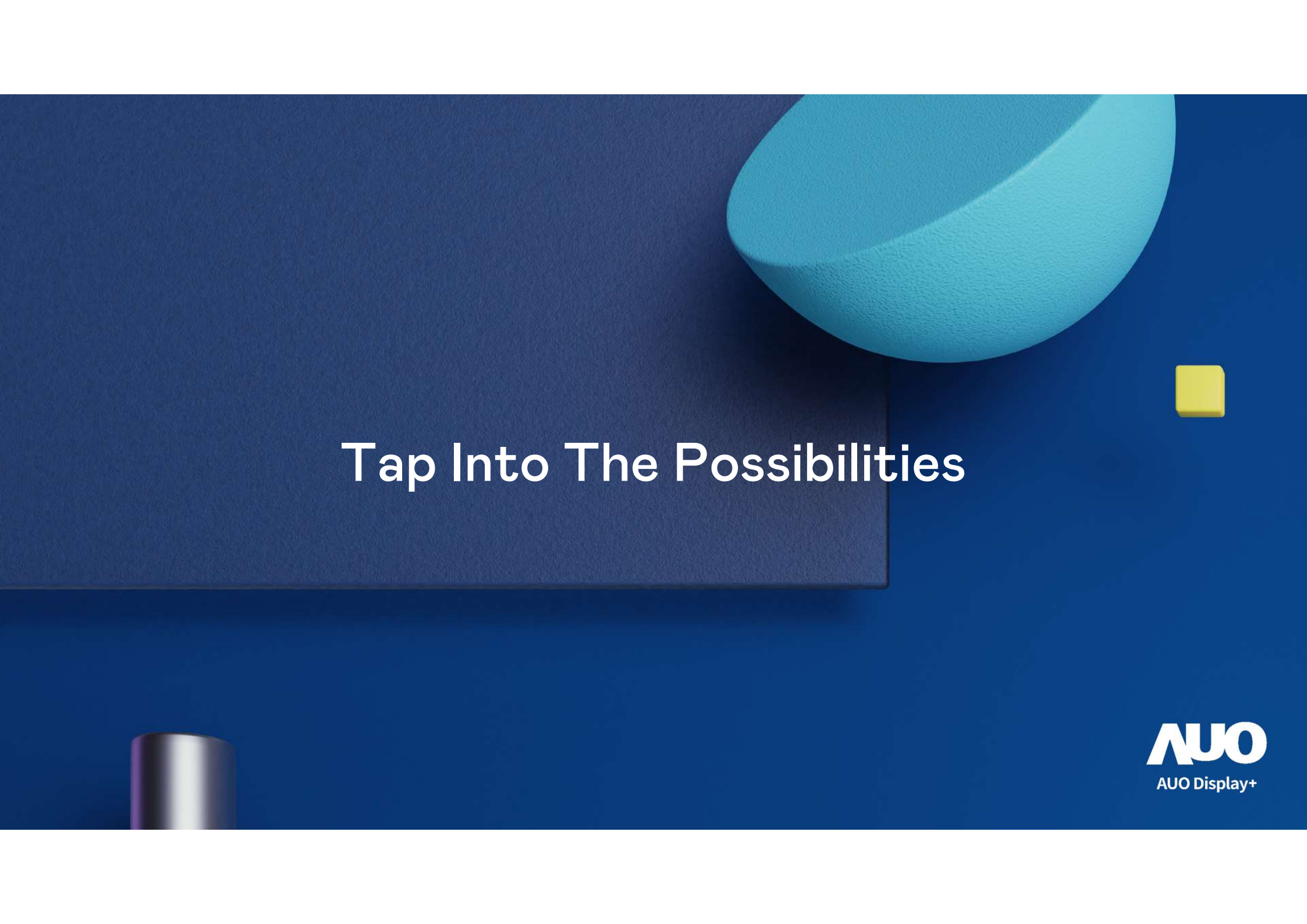


Appendix : Reflector pictures



TORAY (As is)				Solartron (To be)			
Vendor/ Type	Reflector (front side)	Reflector (rear side)	Reflector assembled in BLU	Vendor/ Type	Reflector (front side)	Reflector (rear side)	Reflector assembled in BLU
TORAY/ TCJ18N				Solartron/ EST188P			

- Front side: Toray reflector's coating is more smooth.
- Rear side : Both of Toray and Solartron materials are all pet and shiny, the difference is small.



Tap Into The Possibilities

AUO
AUO Display+